

RAG Question Answering

One-Page Technical Explanation · Design Decisions · Retrieval Evaluation · Observed Failure · Cost Observability

1. Design Parameter — Structure-Aware Chunking

Final configuration: 700-character target chunk size with 120-character overlap.

The chunking pipeline preserves page boundaries, detects section headings, keeps headings with their following content, groups paragraphs into semantic units, and splits oversized units only at sentence or word boundaries. This was selected instead of blind fixed-size splitting because document structure can preserve important retrieval context.

2. Observed Failure — Architecture Retrieval

During development, an architecture-related question initially failed to retrieve the relevant information within the top 3 results; the relevant content appeared at **rank 4**. The observed cause was that the initial chunking approach did not sufficiently preserve the relationship between a section heading and its following content.

Change made: the chunker was redesigned to preserve section structure and keep related heading/content together. In the six-question retrieval experiment, Hit Rate@3 improved from **83.33% (5/6)** to **100% (6/6)**.

3. Evaluation Metric — Retrieval Hit Rate

The retrieval evaluation uses six answerable questions. A question counts as a hit when at least one relevant chunk appears within the retrieved Top-K results.

Top-K	Answerable	Hits	Hit Rate
1	6	4	66.67%
3	6	6	100%
4	6	6	100%

Interpretation: Top-K=1 missed 2 of 6 answerable questions, while Top-K=3 retrieved all 6. Increasing K from 3 to 4 produced no additional hits, so **Top-K=3** was selected as the current retrieval setting. The evaluation set is small and is not intended to be a general RAG benchmark.

Similarity threshold: a fixed distance threshold was considered but not enabled because answerable and unanswerable distance distributions overlapped. For example, an unanswerable question produced a distance of **0.5807**, while an answerable architecture question produced **0.9235**. A simple fixed threshold was therefore unsuitable for the current evaluation.

4. LLM Usage & Estimated Cost Observability

Each query records **prompt tokens**, **completion tokens**, **total tokens**, **cached input tokens** when available, and an **estimated API cost**. These values are logged together with retrieval count and query latency, providing basic inference-cost and performance observability.

The estimated cost is calculated from configured model pricing when pricing information is available. If pricing is unavailable for the selected model, token usage remains available while the cost is reported as unavailable. The reported value is an estimate for observability and should not be treated as an exact billing amount.

5. Unfinished Work & Next Steps

- **Reranking:** Add a cross-encoder or similar reranker after initial vector retrieval.
- **Hybrid retrieval:** Combine dense semantic search with lexical retrieval for exact technical terms.
- **Document understanding:** Add OCR and improved extraction for scanned PDFs, tables, and complex layouts.
- **Evaluation:** Expand the question set and add answer correctness, groundedness, and faithfulness metrics.
- **Production readiness:** Add authentication, multi-user isolation, hosted vector storage, monitoring, and deployment.

Current scope

Local PDF/TXT RAG system with FastAPI, BGE-small embeddings, persistent ChromaDB, Top-K retrieval, Groq grounded generation, source attribution, abstention via **NOT_FOUND**, automated tests, retrieval evaluation, and per-query usage/cost logging.